

IoT-enabled digital product passports and traceability for end-of-life management and circularity of photovoltaic panels across the value chain.

Energy × Product traceability and digital product passport × IoT platform · OS060 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
36 is the world moving	43 can we play, can we win	HIGH 0 named in the evidence	€171m observed evidence · per year	LATER research and standards activi...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity focuses on IoT-enabled digital product passports and traceability for photovoltaic panels, addressing end-of-life management and circularity across the value chain. It targets energy sector stakeholders—manufacturers, recyclers, and asset owners—who face regulatory pressure and operational inefficiencies in managing panel lifecycle data. The opportunity exists now because EU regulations like the Ecodesign for Sustainable Products Regulation mandate digital product passports, and research projects like SOPHIA are already exploring digital solutions for PV panel circularity.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€207m €46.4m – €2.6bn	€171m €38.2m – €2.2bn	€2.1m €459k – €26.0m	observed
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.1m €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting IoT platform	51.7 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

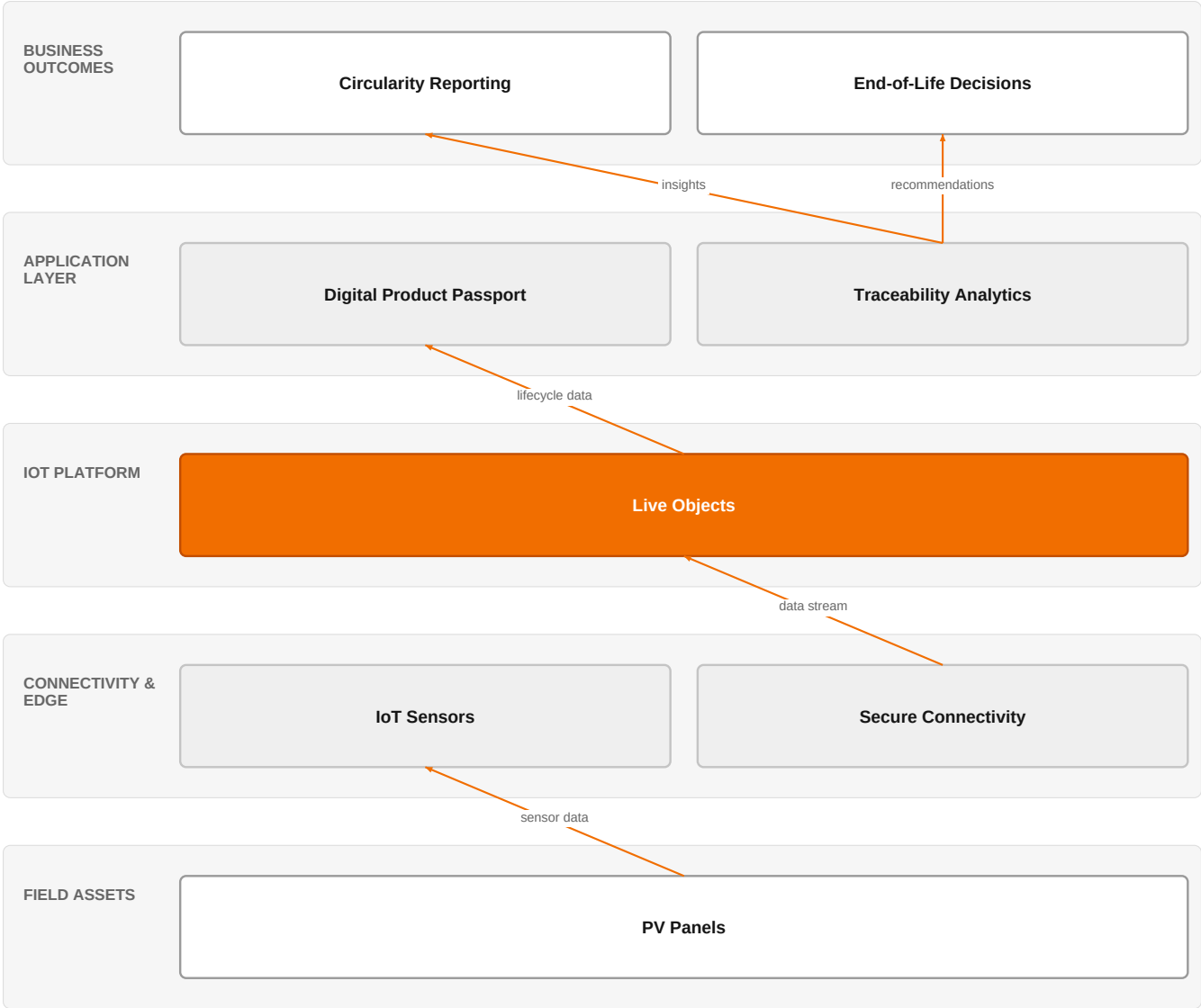
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07

Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a
--	--------------	------------	---------------------------------------	----------------

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

IoT-Enabled PV Panel Digital Passport



This diagram shows how sensor data from PV panels flows through Orange's Live Objects platform to build a digital product passport that supports circularity reporting and end-of-life decisions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulatory mandates are driving the need for digital product passports: the EU's Ecodesign for Sustainable Products Regulation requires DPPs for electronics and appliances by 2027, creating an urgent need for machine-readable lifecycle data. Research highlights that DPPs are emerging as core infrastructure for sustainable manufacturing, enabling lifecycle transparency and circular economy practices. However, significant challenges remain in achieving detailed part-level traceability, managing fragmented supply chain data, and automating compliance—challenges that IoT and digital twins can address. The SOPHIA project specifically targets PV panels, aiming to implement advanced digital solutions involving all value chain actors to increase reuse and recycling.

Sources: SIG-36FFB2435F8A openalex.org 2026-04-20; SIG-F47C31AB60CB MDPI AG 2026-02-14; SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-4FE21FD2ED06 openalex.org 2026-01-01; SIG-A2CC51B21A97 cordis.europa.eu 2025-06-01

Who buys, and why now

The buyer is typically a sustainability or operations executive at a PV manufacturer, a large-scale solar asset owner, or a specialized recycler. The pain is operational: they lack end-to-end visibility into panel composition, condition, and location, making end-of-life decisions inefficient and compliance reporting manual. The trigger is often an upcoming regulatory deadline (e.g., ESPR) or a need to demonstrate circularity to investors or customers. For recyclers, the pain is the inability to sort and recover materials efficiently without accurate product data.

Sources: SIG-BA196FF05244 Springer Science and B 2026-02-03; SIG-4FE21FD2ED06 openalex.org 2026-01-01

Why it is hot now — every claim bound to a dated source

- The SOPHIA project aims to implement advanced digital solutions involving all actors in the value chain to increase the reuse and recycling of photovoltaic panels. [SIG-A2CC51B21A97]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using Live Objects as the IoT platform to collect and manage sensor data from PV panels (e.g., location, performance, environmental conditions). IoT experts would design and deploy edge sensors and connectivity, while Orange Group researchers would contribute to data modeling and interoperability standards. The engagement would start with a pilot to capture panel data, then build a digital product passport data layer, and finally integrate with customer systems for compliance reporting and circularity analytics. Certifications like ISO 27001 and SecNumCloud would underpin data security and sovereignty.

Why Orange can win

Orange can win by leveraging its certified, sovereign infrastructure (ISO 27001, SecNumCloud) to handle sensitive lifecycle data, which is a differentiator in the EU regulatory context. Live Objects provides a scalable IoT platform, and Orange's research capability adds credibility in a nascent field. However, the asset list is thin—there is no dedicated DPP or traceability offering, so Orange would need to partner or build custom solutions, which could slow time-to-market.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

HIGH competitive intensity (score 8.2; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitors include hyperscalers like AWS, telecom peers like Deutsche Telekom, Telefonica Tech, and Vodafone Business, and global integrators like NTT DATA. AWS is strong in cloud scale and IoT platform breadth; Deutsche Telekom and Vodafone have extensive IoT connectivity and platform offerings; NTT DATA brings systems integration expertise. Telefonica Tech is active in the energy sector. The customer will compare on platform maturity, industry-specific solutions, and the ability to meet regulatory compliance—areas where Orange's sovereign credentials and research depth could stand out, but competitors have more mature IoT ecosystems.

Sources: SIG-A2CC51B21A97 cordis.europa.eu 2025-06-01

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells IoT platform; competes in OX: Smart Industries — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—

NTT DATA	Global systems integrator	sells IoT platform; competes in OX: Smart Industries	structural	—
Proximus	Telecom operator peer	sells IoT platform	structural	—
Telefonica Tech	Telecom operator peer	sells IoT platform; active in Energy	structural	—
Telia Company	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—
Verizon Business	Telecom operator peer	sells IoT platform	structural	—
Vodafone Business	Telecom operator peer	sells IoT platform; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently track the composition and condition of your PV panels at end-of-life?
- 2 What regulatory or customer pressure are you facing to demonstrate circularity or provide product passports?
- 3 Do you have existing IoT sensors or connectivity on your panels, or would this be a greenfield deployment?
- 4 Who in your organization owns the data from panel lifecycle, and how is it shared with recyclers or regulators?
- 5 What is your biggest operational pain in end-of-life management: data collection, compliance reporting, or material recovery efficiency?

Objections you will meet

They will say	You can answer
We already have a basic tracking system for our panels; why do we need a digital product passport?	A basic system may track location and performance, but a DPP goes further by capturing full lifecycle data—materials, repairs, and end-of-life status—in a standardized, machine-readable format. This is what regulators are moving toward, and it enables better circularity decisions, like identifying panels suitable for reuse or efficient recycling.
The regulatory mandate for PV panels isn't clear yet; why invest now?	You're right that the specific mandate for PV is not yet final, but the ESPR sets the direction, and early movers can shape standards and gain a competitive advantage. Also, the operational benefits—reduced manual reporting, better material recovery, and improved asset management—are valuable regardless of regulation.
We're concerned about data security and sovereignty when sharing lifecycle data across the value chain.	That's a valid concern. Orange's infrastructure is certified to ISO 27001 and SecNumCloud, which are recognized for high security and sovereignty. We can design the solution to give you control over data access, ensuring compliance with EU data protection requirements.

Risks and unknowns

The main risk is regulatory timing: DPP mandates for PV panels are not yet explicit, and the market may be waiting for clearer standards. Technical challenges include achieving granular traceability across fragmented supply chains and ensuring data interoperability. The SOPHIA project is still research-stage, so proven reference implementations are lacking. Additionally, Orange's asset list is limited, and competitors may already have more comprehensive IoT and DPP solutions, making differentiation harder.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Assemble a bid angle on Product traceability and digital product passport with IoT platform, combining Live Objects (L1).
Sales	Open with Energy accounts on Product traceability and digital product passport, leading with Live Objects (L1) — confirm an internal proof point before the meeting.
Strategist / Innovator	Commission a deep-dive on Product traceability and digital product passport in Energy using IoT platform: size the opportunity bottom-up and decide whether to prototype next quarter.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-36FFB2435F8A	2026-04-20	openalex.org	1	A Comprehensive Review of Digital Product Passports in Sustainable Manufacturing: Current State...
SIG-A3F10F84C60F	2026-03-01	APL Electronic Devices	1	Industry perspective: Wearable electronic devices for remote patient monitoring
SIG-F47C31AB60CB	2026-02-14	MDPI AG	1	Dynamic Traceability and Reliability Management: Integrating the Digital Product Passport with ...
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-4FE21FD2ED06	2026-01-01	openalex.org	1	On the Utilization of Digital Product Passports for Intelligent Metal Sorting and Recycling
SIG-DFE69DA138F9	2026-01-01	openalex.org	1	Shaping a circular future for electronics: The role of Digital Product Passports
SIG-6786E4AD1713	2025-11-22	openalex.org	1	The quest for more circular battery value chains: Implementing the EU digital battery passport ...
SIG-BCCCB520B0E1	2025-10-21	Institute of Electrical a...	1	Design and Integration of IoT-Enabled Health Sensors for Remote Patient Monitoring
SIG-EC1CEFFD337A	2025-10-15	openalex.org	1	Digital Product Passport in support of Data-Driven End-of-Use Strategies—a systems design persp...
SIG-EFEBD520406D	2025-10-15	openalex.org	1	Stakeholder perspectives on digital product passports for construction products
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-A2CC51B21A97	2025-06-01	cordis.europa.eu	1	Implementation of Advanced Digital Solutions to increase the circularity of PV panels throughou...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:44:04+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.